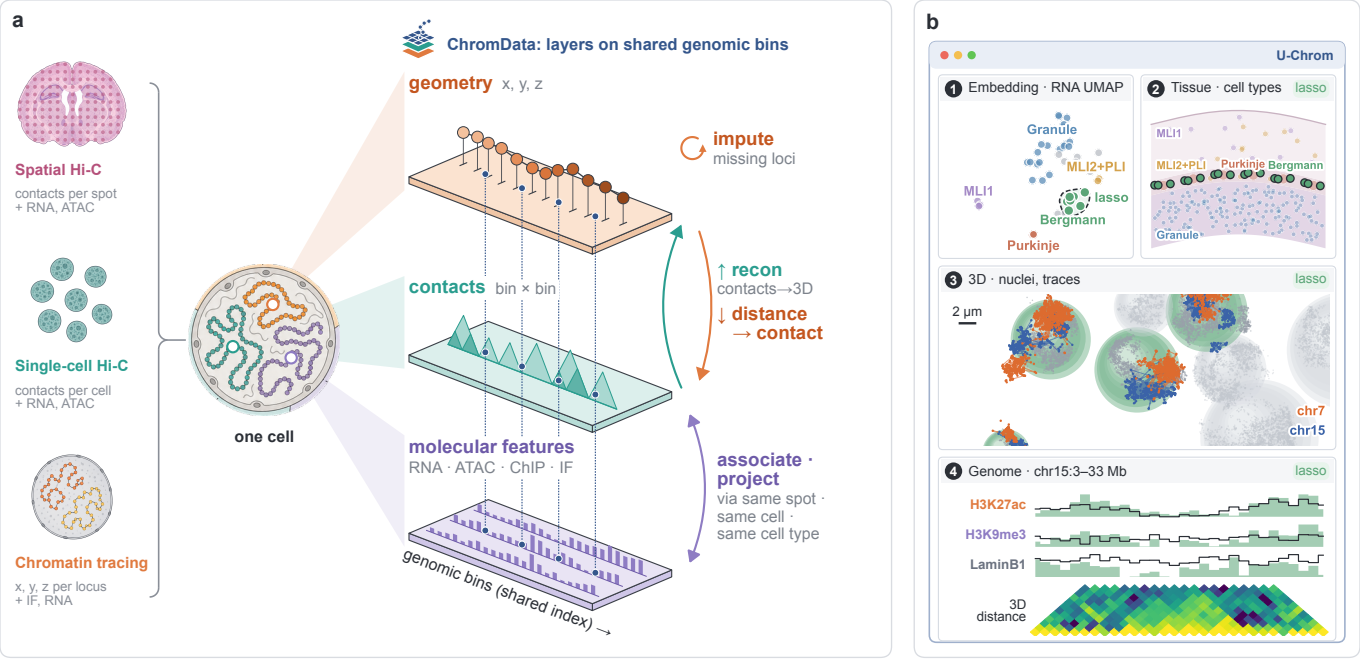

a
ChromData: layers on shared genomic bins
Spatial Hi-C
contacts per spot
+ RNA, ATAC
Single-cell Hi-C
contacts per cell
+ RNA, ATAC
Chromatin tracing
x, y, z per locus
+ IF, RNA
one cell
geometry x, y, z
impute
missing loci
contacts bin × bin
↑ recon
contacts→3D
↓ distance
→ contact
molecular features
RNA · ATAC · ChIP · IF
associate · project
via same spot ·
same cell ·
same cell type
genomic bins (shared index) →
b
U-Chrom
1 Embedding · RNA UMAP
Granule
MLI2+PLI
MLI1
lasso
Bergmann
Purkinje
2 Tissue · cell types lasso
MLI1
MLI2+PLI
Purkinje
Bergmann
Granule
3 3D · nuclei, traces lasso
2 μm
chr7
chr15
4 Genome · chr15:3–33 Mb lasso
H3K27ac
H3K9me3
LaminB1
3D distance
c
Interfaces
people · agents
Python API
web browser, local or hosted
atlas: 22 datasets, 11 studies
WebMCP + agent skills
Analysis
uchrom, one API
uchrom.recon
3D structures from contacts
uchrom.im
spots → traces; missing loci
uchrom.strc
TADs, loops, compartments
uchrom.fea
per-locus and per-cell features
uchrom.emb
embeddings, clusters
uchrom.spatial
spatial tests, stripes removed
Formats
& standards
4DN
FOF-CT
Juicer
.hic
.pairs
.cool · .mcool · .scool
pyHiM
SnapFISH
structures
.3dg
read
ChromData
.chromdata.zarr
write · link
Zarr
Parquet
4DN
FOF-CT
scverse
AnnData
MuData
SpatialData
.mcool
.scool
structures
PDB → PyMOL / ChimeraX